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SOFTWARE PROJECT

FoodieNetwork

Cooking Lovers Application

Chosen Technology Stack

For our Cooking Lovers Application, we decided to work with React and Firebase frameworks for the Backend. Along with HTML and CSS for the Frotnend to help with making the platform look more appealing and user-friendly.

Backend Technologies:-

-React

React(ReactJS/React.js) is a free and open-source front-end JavaScript library that aims to make building user interfaces based on components more "seamless"

Because React is only concerned with the user interface and rendering components to the DOM, React applications often rely on libraries for routing and other client-side functionality. A key advantage of React is that it only re-renders those parts of the page that have changed, avoiding unnecessary re-rendering of unchanged DOM elements.

Key Features:

- **Component-Based** – UI is built using reusable components.
- **Virtual DOM** – Efficient updates by comparing and modifying only changed parts.
- **Declarative UI** – Simplifies UI development by describing how the UI should look based on state.
- **State Management** – Uses `useState` and `useReducer` for managing component state.
- **Props** – Pass data between components easily.
- **Hooks** – Enable functional components to use state and lifecycle features (`useEffect`, `useContext`, etc.).
- **One-Way Data Binding** – Ensures predictable data flow.
- **Fast Rendering** – Uses reconciliation to optimize performance.
- **React Router** – Enables navigation in single-page applications (SPA).
- **Rich Ecosystem** – Works well with libraries like Redux, Next.js, and TailwindCSS.

React's component-based structure, efficient rendering, and state management make it the perfect choice for building FoodieNetwork, ensuring a seamless, interactive, and scalable user experience.

-Firebase

The Firebase Realtime Database is a cloud-hosted database. Data is stored as JSON and synchronized in realtime to every connected client.

We chose Firebase because when you build cross-platform apps with JavaScript Software Development Kits, all of your clients share one Realtime Database instance and automatically receive updates with the newest data, which helps in achieving our goal with this app.

Key Features:

- **Authentication** – Easy user login with email/password, Google, Facebook, etc.
- **Firestore Database** – Real-time NoSQL cloud database for storing and syncing data.
- **Cloud Storage** – Store and manage images, videos, and files securely.
- **Hosting** – Fast and secure hosting for web apps with free SSL.
- **Cloud Functions** – Run backend logic without managing servers.
- **Push Notifications** – Send real-time notifications to users.
- **Analytics** – Track user behavior and app performance.
- **Machine Learning** – Built-in AI tools for personalization and automation.

Firebase works as a backend-as-a-service (BaaS**)**, handling authentication, databases, storage, and hosting without the need for a dedicated backend server. It syncs data in real-time with Firestore, manages user logins securely, and enables cloud functions for automated processes. This makes it ideal for dynamic apps like FoodieNetwork, ensuring smooth data management and scalability.

Frontend Technologies:-

HTML

Hypertext Markup Language (HTML) is the standard markup language for documents designed to be displayed in a web browser. It defines the content and structure of web content. It is often assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript, a programming language.

Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for its appearance.

HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes, and other items.

Key Features:

- **Structure** – Defines the backbone of web pages using elements like `<header>`, `<section>`, and `<footer>`.
- **Semantic Elements** – Improves readability and SEO with tags like `<article>`, `<nav>`, and `<aside>`.
- **Forms & Inputs** – Enables user interaction through `<form>`, `<input>`, `<button>`, etc.
- **Multimedia Support** – Embeds images, audio, and videos using `<img>`, `<audio>`, and `<video>`.
- **Hyperlinks & Navigation** – Connects pages via `<a>` tags.
- **Tables & Lists** – Organizes content using `<table>`, `<ul>`, and `<ol>`.
- **Responsive Design** – Works with CSS and media queries for mobile-friendly layouts.

HTML provides the **structure** of a webpage, defining content elements that browsers render visually. Combined with CSS for styling and JavaScript(React and Firebase) for interactivity, it forms the foundation of web development.

CSS

HTML is best paired with CSS, which is why we chose the pair. Cascading Style Sheets (CSS) is a style sheet language used for specifying the presentation and styling of a document written in a markup language such as HTML or XML (including XML dialects such as SVG, MathML or XHTML) CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript.

CSS is designed to enable the separation of content and presentation, including layout, colors, and fonts. This separation can improve content accessibility, since the content can be written without concern for its presentation; provide more flexibility and control in the specification of presentation characteristics; enable multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, which reduces complexity and repetition in the structural content; and enable the .css file to be cached to improve the page load speed between the pages that share the file and its formatting.

Key Features:

- **Styling & Layout** – Controls colors, fonts, spacing, and positioning.
- **Selectors & Specificity** – Targets elements efficiently (`.class`, `#id`, `element`).
- **Box Model** – Defines margins, borders, padding, and content.
- **Flexbox & Grid** – Enables responsive and flexible layouts.
- **Media Queries** – Adapts designs for different screen sizes.
- **Animations & Transitions** – Adds motion effects for better UI experience.
- **Variables & Custom Properties** – Allows reusable styles for consistency.

CSS styles HTML elements to create visually appealing web pages. It works by selecting elements and applying properties like colors, sizes, and positioning, ensuring a structured and responsive design.

Application Architecure

Application architecture is the process of defining a structure that meets all the technical and operational requirements of an application. It’s like a blueprint that shows the design of a system and portrays how its’ components work together to achieve an objective.

We have chosen *FoodieNetwork* as the name for our application.

Frontend:

Using HTML and CSS for the User Interface to enable interactivity with the app, the rendering of the webpages such as login, sign up, profiles, the feed, and post recipes.

-Data like recipe details will be displayed, along with community engagement like comments and likes, as well as recipe sharing. This data is sent to the backend side of the application.

Backend:

React and firebase were the chosen frameworks to aid in building the backbone of the app, allowing us to process the user’s requests(engagement, sharing, liking, commenting, etc) and manage the data.

-Anything related with making the application actually function like any other social media platform is handled in the backend.

Data Flow:

-The frontend is used for user interaction, while the backend receives the data through form submission or using HTTP requeests, and then handles and processes it.

-The backend will once again transfer the data it processed back to the frontend, updating the User Interface(number of likes, new recipe posted, comments, etc),

-Content uploaded by the user will be store in the media folder and managed by Firebase.

Description of Application Functionality

Our app will include a login page for pre-existing registered users, alternatively, new users can sign up and create an account.

On the user's feed, followed accounts and their posted content along with recipes will be hosted there. A popular dishes section will also be included in the homepage(feed).

The search bar included will allow users to search for almost anything within the app, from other users, to recipes, and generated grocery lists.

Users will be able to like, comment and share other's posts/recipes, or their own content if needed.

Saving a recipe for later is also an added feature.

Shopping lists can be generated based on the user's wanted recipe.

Account management will be included, deleting posts or account, managing other user's, along with profile pictures and security measures.

User Interface Description

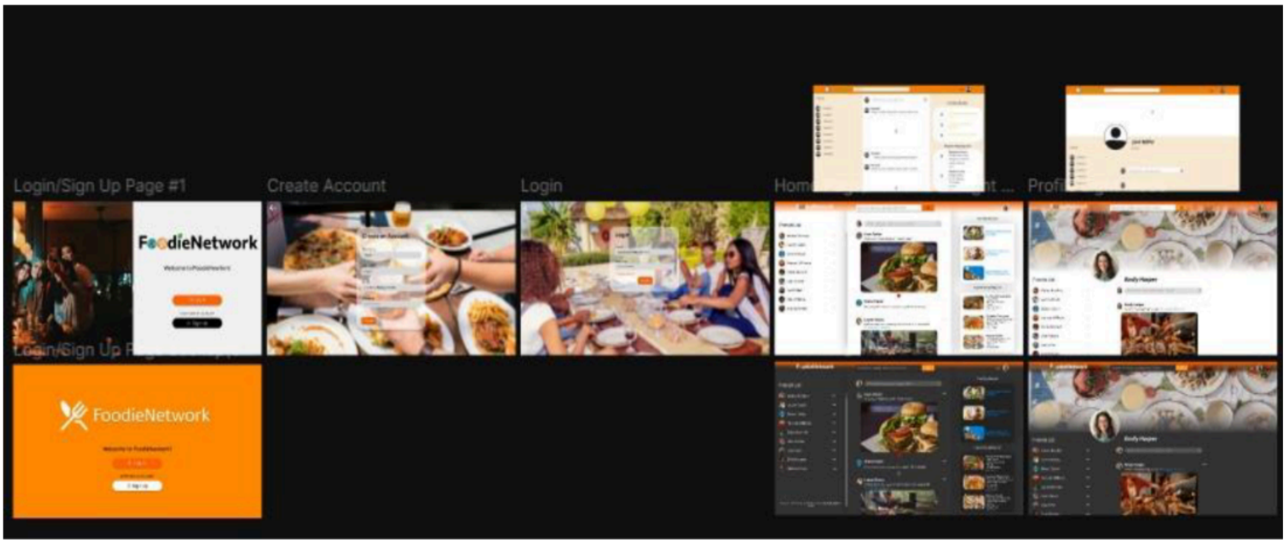
The chosen primary color for our application is orange, which is an energetic color commonly involved with freshness and cuisine, giving the user a feeling of warmth, excitement, and in the right place. A clean and modern design while maintaining noise and loud to keep the users intrigued.

An abundance of buttons and icons were provided to make the User Experience enjoyable and easy.

Pages Included:-

- Login - enabling users to access their accoutns
- Sign Up - new users will also have the possibility of engaging with our app
- Homepage - everything will be found here, access to profile, friends, posts, trending recipes, and anything else the user might need
- Profile - account management, deleting account, removing followers, user's shared content, personal details

With time, we hope to make our app accessible on mobile phones.



Summary

Our app, as the name indicates, is for cooking lovers and enthusiasts hoping to connect and share their passion for cuisine with the world and individuals of similar interests.

HTML, CSS, are primarily used for the frontend due to ease of navigation and flexibility.

Figma was used to design what the app should look like in its final form.

React and Firebase were the most suitable options for our backend side of the app because it fulfills our needs, being a versatile and powerful framework without adding complication to usage.

We hope to add more features in the future to make this app as functional as possible.

Sources:-

wikipedia, figma, w3schools.com, firebase.google.com